

ParcelPilot Configuration and Environment Reference

Service URLs, paths, timeouts, thresholds, snapshots, and safe checks.

ParcelPilot operations reference

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1. Configuration overview

ParcelPilot uses environment-style values for service URLs, local database paths, snapshot locations, timeouts, and thresholds.

Checks

- Confirm the key exists when required.
- Confirm the value has the expected type or path form.
- For file paths, check existence and readability.
- Keep secret values masked.

Interpretation

Configuration evidence should be combined with the route symptom and service logs. A changed value that belongs to an unrelated service is not a useful explanation.

After a change

Record whether the service was restarted or reloaded and rerun the original failing request.

Operational detail

Configuration should be read in the context of the process that consumed it. A snapshot taken after a change does not prove the running service has reloaded the new value. Record the nearest restart or reload event where the training logs provide one.

For path settings, check the chain in order: the key exists, the path is syntactically usable, the referenced file or directory exists, and the service has the required access. Skipping directly from 'present' to 'healthy' can miss permission or content problems.

Example review

- Compare the current value with the previous snapshot.
- Check whether the key belongs to the affected service.
- Look for a startup or health message after the change.
- Rerun the failing route after correction.

A configuration change on an unrelated service should remain in the incident timeline if it occurred nearby, but it should not be promoted to root cause without matching evidence.

2. API service configuration

AUTH_SERVICE_URL, ORDER_SERVICE_URL, TRACKING_SERVICE_URL, DOCUMENT_SERVICE_URL, NOTIFICATION_SERVICE_URL, and REPORTING_SERVICE_URL identify downstream services.

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Normal operating pattern

A normal API request has a request_id, a route, a downstream call when needed, and a final status. Basic health normally completes in tens of milliseconds, while an order route includes the time spent waiting for the orders service.

Failure pattern

A generic API-500 is most useful when correlated with a downstream code on the same request_id. If the downstream service returns a specific error, keep that code visible in the incident record instead of treating API-500 as the final diagnosis.

Worked example

For req-0048123, API-500 at 14:06:18 followed an ORD-500 from the orders service under the same request identifier. That pattern points to the orders path rather than the API process itself.

Useful evidence to keep

- The exact incident time or best available time window.
- The route, service, and request_id when available.
- The specific error code rather than only the HTTP status.
- One known-good comparison case when practical.

The operator should also note what was checked and found to be normal. This prevents the next person from repeating the same work and helps keep an unrelated warning from becoming the assumed cause.

3. Orders configuration

ORDERS_DB_PATH identifies the SQLite database. ORDERS_SCHEMA_VERSION records the application version expected by the service. DB_BUSY_TIMEOUT_MS controls selected lock waits.

Checks

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Interpretation

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After a change

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Normal operating pattern

Normal order lookup reads the configured SQLite file, finds the requested order, and returns order details independently of tracking. ORD-404 for an unknown identifier is expected application behavior.

Failure pattern

ORD-500 DB_OPEN_FAILED points to the database path or file access. ORD-409 points to schema version mismatch. These failures can both become HTTP 500 responses at the API layer, but the corrective checks are different.

Worked example

If ORDERS_DB_PATH is missing and every tested order ID returns ORD-500, the missing database path is stronger evidence than a simultaneous tracking warning on another request.

Useful evidence to keep

- The exact incident time or best available time window.
- The route, service, and request_id when available.
- The specific error code rather than only the HTTP status.
- One known-good comparison case when practical.

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4. Tracking configuration

CARRIER_SNAPSHOT_PATH identifies carrier data. CARRIER_STALE_AFTER_MINUTES defines freshness. TRACKING_TIMEOUT_MS defines the timeout.

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After a change

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Normal operating pattern

Normal tracking reads a recent carrier snapshot and returns a sequence of shipment events. Some orders legitimately have fewer events than others, so the number of events alone is not an error.

Failure pattern

TRK-206 means the response is incomplete, TRK-409 means the snapshot is too old, TRK-422 means the source cannot be used safely, and TRK-504 means the carrier lookup exceeded its timeout.

Worked example

A snapshot updated eight minutes ago with two events for one order may be valid partial data. A snapshot updated two hours ago should be treated as stale even if the JSON structure is valid.

Useful evidence to keep

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5. Documents configuration

DOCUMENT_STORE_PATH identifies shipment files. DOCUMENT_MAX_FILE_MB is an operational file-size setting.

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After a change

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Normal operating pattern

Document routes first confirm the order and then look for the requested file under the configured store. Some orders do not have proof-of-delivery or customs documents, so a controlled not-found response can be normal.

Failure pattern

DOC-404, DOC-403, and DOC-500 should remain separate. A missing file, an unreadable file, and an invalid document root have different causes and different next checks.

Worked example

If label.pdf opens but proof.pdf returns DOC-404, do not change the global document-store path. Verify whether proof.pdf exists for that order.

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6. Notifications configuration

NOTIFICATION_QUEUE_PATH points to queue data. QUEUE_BACKLOG_THRESHOLD controls the backlog warning. Provider credentials are secret and remain masked.

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For path settings, check the chain in order: the key exists, the path is syntactically usable, the referenced file or directory exists, and the service has the required access. Skipping directly from 'present' to 'healthy' can miss permission or content problems.

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7. Reporting configuration

REPORTING_DB_PATH identifies the analytics database. REPORT_QUERY_TIMEOUT_MS controls timeout. MAX_REPORT_DAYS validates date ranges.

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Normal operating pattern

Reporting reads a separate analytics database. Small date ranges normally complete faster than large ones, and the same report can return different row counts as the requested window changes.

Failure pattern

RPT-422 is a validation problem, RPT-500 is a query failure, and RPT-504 is a timeout after execution begins. A timeout should be investigated with date range, query type, and repeated duration.

Worked example

A 90-day report timing out while 7-day reports complete in 300 ms is not the same as every report failing immediately with RPT-500.

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8. Timeouts and thresholds

Timeouts prevent indefinite waiting. Thresholds mark conditions such as stale data or queue backlog. Increasing them is not automatically a fix.

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9. Configuration snapshots

Snapshots record values at `captured_at`. They do not prove a running service reloaded a changed value without a restart.

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10. File path checks

For local paths, distinguish missing setting, missing file, unreadable file, and wrong content. These conditions map to different errors.

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11. Secrets and safe output

Tools can report present, missing, masked, valid, unreadable, or unreachable. They should not return secret values.

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12. Change review

When an incident follows a configuration change, compare before and after snapshots and verify that the changed key belongs to the affected request path.

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